Classifying Parliamentary Text into Ministries

A Study of Representation Methods

DS5601: Natural Language Processing

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Problem Motivation

- Parliamentary proceedings are released as long, unstructured documents.
- Important information (topics, ministries) is not explicitly labeled.
- **Goal:** Identify which ministry a paragraph belongs to.
- **Key Challenge:**
    - No annotated dataset
    - High domain complexity
- **Core Idea:**
    - Explore different paradigms of text representation
    - Evaluate methods without relying on gold labels

Data and Setup

- **Data:**
  - Parliamentary transcripts (debates, budget speeches, Q&A)
  - 53 ministry knowledge bases (Wikipedia + Govt + LLM summaries)
- **Processing:**
  - Speaker-wise and paragraph-level segmentation
  - Cleaning + domain-specific stopwords
- **Evaluation:**
  - No ground truth labels
  - Weak supervision using LLM + cross-method comparison

Topic Modeling (Supporting Structure)

- Topic modeling performed using **David Mimno's online tool** `https://mimno.infosci.cornell.edu/jsLDA/`
- **Outputs obtained:**
  - 25 topics over the full corpus
  - Top 10 representative words per topic
  - Document–Word–Topic state distributions
  - Automatically generated domain-specific stopwords

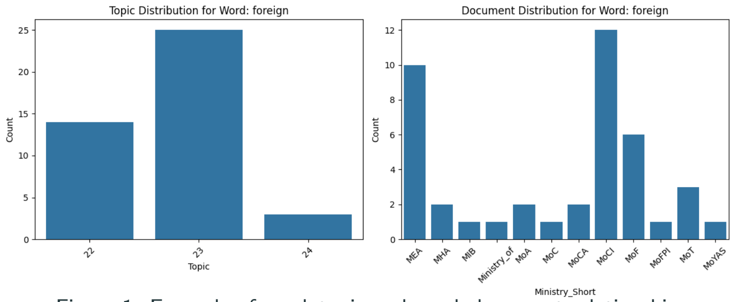

Figure 1: Example of word–topic and word–document relationships

Three Ways to Represent Text

Method	Core Idea
Lexical (TF-IDF)	Meaning = Words
Probabilistic	Words → Topics → Ministries
Embedding-based	Meaning = Vector similarity

Research Question: Which representation works best when no labels are available?

Method 1: Lexical (TF-IDF)

Idea: Match paragraph words with ministry-specific vocabulary

Assumption:

- Words directly indicate meaning

Works well when:

- Distinct domain keywords exist (e.g., "fertilizer", "crop")

Fails when:

- Synonyms are used ("budget" vs "fiscal policy")
- Vocabulary overlaps across ministries

Insight: Parliamentary language is not discriminative enough for pure keyword methods.

Method 2: Probabilistic (Doc-Word-Topic)

Idea: Words $\rightarrow$ Topics $\rightarrow$ Ministries

$Score(d) = \sum_w \sum_t P(d|w) \cdot P(t|w) \cdot P(d|t)$

Assumptions:

- Words are independent
- Topics explain word usage
- Frequencies approximate probabilities

Strength: Combines corpus statistics with structure

Fails because:

- Independence assumption is violated
- Topics overlap across ministries

Insight: Real language does not follow clean probabilistic assumptions.

Method 3: Embedding-Based

Idea: Represent text as vectors and compare using similarity

Variants:

- Whole-document embedding (loses detail)
- Mean of paragraph embeddings (**best**)
- Top-K similar paragraphs (captures local similarity)

Why it works:

- Captures semantic meaning beyond keywords
- Handles synonyms and paraphrases

Limitations:

- Sensitive to representation design
- Can blur multiple topics

Embedding Space Visualization

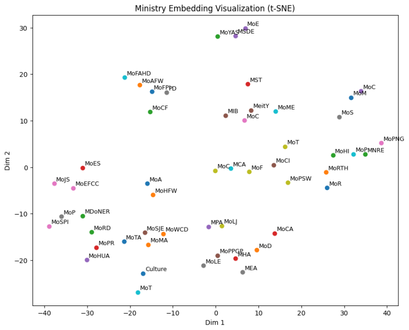

Figure 2: t-SNE projection

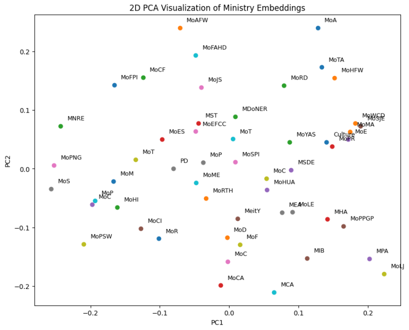

Figure 3: PCA projection

Observations:

- Some ministries form clear clusters
- Significant overlap exists between related ministries

Clustering Ministries

- Observed weak separation (low silhouette scores) for hard clustering
- Adopted **consensus-based clustering**: Group ministries based on stable co-occurrence patterns

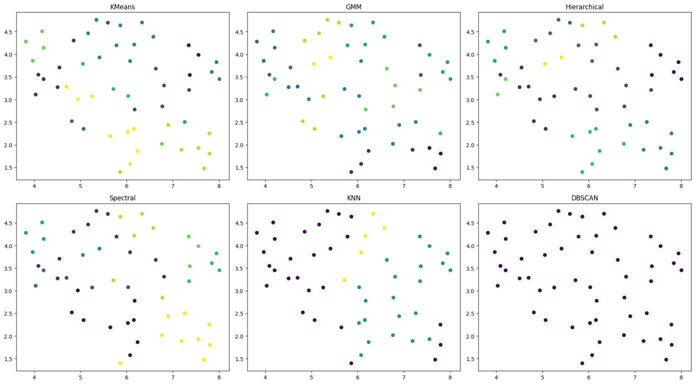

Figure 4: Consensus-based grouping of ministries

Comparison with Weak Supervision (ChatGPT Labels)

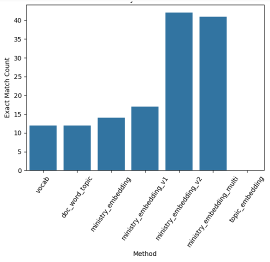

Figure 5: Exact Match

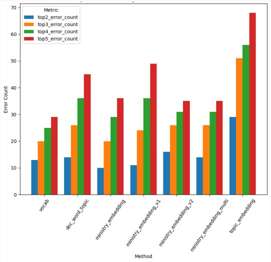

Figure 6: Top-K Errors

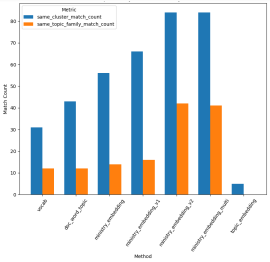

Figure 7: Cluster Match

Observations:

- Embedding methods show higher consistency across metrics
- Probabilistic method improves over lexical baseline
- Exact match is low, but cluster-level agreement is higher

Insight: In absence of ground truth, consistency across views is more meaningful than raw accuracy.

Results and Key Observations

- Embedding-based methods outperform lexical approaches
- Probabilistic method improves over simple keyword matching
- Best performance: **paragraph-level embedding averaging**

Important Observation:

- No method is perfectly correct
- Some methods are more **consistent** than others

Reason:

- High semantic overlap between ministries

Conclusion and Takeaways

- Ministry classification is fundamentally a **representation problem**
- **Lexical methods:**
    - Fail due to vocabulary overlap and lack of context
- **Probabilistic methods:**
    - Limited by unrealistic independence and topic assumptions
- **Embedding methods:**
    - Capture semantics effectively
    - Performance depends on representation design (granularity)

Key Insight: Better models did not solve the problem — better representations did.

Thank You

Questions?

Project Repository:
https://github.com/DevadattaP/analyse_parliament_transcripts_NLP